# Curriculum Vitae


*Name:* **Santosh Kumar Panda**, Prof. of GIScience

*Address:* 2150 N. Koyukuk Drive (O'Neill 368)
University of Alaska Fairbanks, Fairbanks, AK 99775
Phone: 907-474-7539 (Off.), 907-371-6615 (Cell)
E-mail: skpanda@alaska.edu


*Education:* **Ph.D., Geology,** 2011, University of Alaska Fairbanks, Alaska
Dissertation: *Permafrost Distribution Mapping and Temperature Modeling along the Alaska Highway corridor,* Advisor: Dr. Anupma Prakash

**M.Sc., Applied Geology**, 2003, Indian Institute of Technology Roorkee, India
Dissertation: *Sequence stratigraphy in shallow marine deposits: a review of late Cretaceous-Eocene sequences of India,* Advisor: Dr. Sunil Bajpai

**B.Sc., Geology and Mineralogy**, 2001, Sambalpur University, India (1st Rank in the University in the Geology Major)

*Biographical Summary:* I received my undergraduate academic training in India, and graduate academic training in India and the U.S.A. My current research is focused on advancing scientific understanding of the climate induced landscape change and related geo-hazards in Alaska such as changes due to large forest fires, thawing permafrost, and eroding riverbanks with an ultimate goal to serve various stakeholders and influence policy decisions in support of sustainable solutions to emerging environmental problems. I use a variety of research techniques that range from detailed field observations to regional scale spatial modeling. My notable research contributions include first-ever high-resolution modeling of permafrost (i.e., perennially frozen ground) distributions in the seven national park units of Alaska, building community capacity to monitor permafrost in the native villages of Telida and Nikolai, mapping of Colville River bathymetry to aid safe subsistence travel, and production of an updated wildfire fuel map (at 10 m) of boreal Alaska. Currently, I am leading the research effort to map statewide forest health and disturbance as part of the USDA NIFA project. Sharing of academic knowledge and research findings with students, colleagues, and the public at large is a high priority for me. I regularly participate in STEAM outreach events and opportunities for public engagement. I have contributed to numerous science outreach efforts in Alaska ranging from speaking to K-12 students in classrooms and at the Permafrost Research Tunnel (located in Fox) to Alaskans across the state including outreach at 21 rural communities.

I teach GIS and remote sensing classes with a focus on natural resource applications at the *Department of Natural Resources and Environment*, *University of Alaska Fairbanks* (UAF).

## WORK HISTORY

2025 - present: Professor of GIScience, Dept. of Natural Resources and Environment, UAF
2020 - 2025: Associate Professor of GIScience, Dept. of Natural Resources and Environment, UAF
2016 - 2020: Assistant Research Professor, Geophysical Institute, UAF
2014 - 2016: Research Associate, Geophysical Institute, UAF (Advisor: Dr. V. Romanovsky)
2011 - 2014: Postdoctoral Fellow, Geophysical Institute, UAF (Advisor: Dr. V. Romanovsky)

I started my current Associated Professor position in fall 2020. I hold a joint appointment with 70% teaching in the College of Natural Sciences and Mathematics and 30% research in the Institute of Agriculture, Natural Resources and Extension (IANRE) during the 9-month academic year and 100% research in summer.

My workload during AY 2020-2021 to AY 2023-2024 including summer is on average 44% teaching (15 – 22 units), 46% research (18 – 21 units) and 10% service (4 units).

## TEACHING

***Associate Professor**, Dept. of Natural Resources and Environment, UAF (Fall 2020 – present)*
***Cooperating Faculty**, Dept. of Geosciences, UAF* (Fall 2015 – present)

I started formal teaching as a postdoc in fall 2015 when I taught a graduate course (GEOS 622: *Digital Image Processing in Geosciences)* for the Department of Geosciences. I joined the Department of Natural Resources and Environment in fall 2020 at the rank of Associate Professor. My current teaching load is 3 courses a year (1 fall + 2 spring). The table below lists all the courses I have taught at UAF and their student evaluation scores.

**List of courses taught at UAF**:

| ***Course No.*** | ***Course Type: Course Name (Credits)*** | ***Evaluation Score*** |
|---|---|---|
| NRM F338 | UG: Introduction to GIS (3) | Fall 2020 (4.6 / 5.0)<br>Fall 2021 (4.2 / 5.0)<br>Fall 2022 (4.3 / 5.0)<br>Fall 2023 (4.7 / 5.0)<br>Fall 2025 ( / 5.0) |
| NRM F435 | UG: GIS Analysis (4) | Spring 2020 (4.4 / 5.0)<br>Spring 2021 (4.8 / 5.0)<br>Spring 2022 (4.5 / 5.0)<br>Spring 2023 (4.7 / 5.0)<br>Spring 2024 (4.6 / 5.0) |
| NRM F638 | G: GIS Programming (3) | Spring 2023 (4.3 / 5.0)<br>Fall 2025 ( / 5.0) |
| NRM F641 | G: Natural Resource Applications of Remote Sensing (3) | Spring 2022 (4.7 / 5.0)<br>*Spring 2024 ( - / 5.0) |
| NRM F697 | G: GIS Applications (3) | Fall 2022 (4.4 / 5.0)<br>*Fall 2023 ( - / 5.0) |
| NRM F697 | G: Seminar: Geospatial Applications in NRE (2) | Fall 2022 (4.4 / 5.0) |
| NRM F697 | G: GIS Analysis in ArcGIS Pro (3) | Fall 2021 (4.4 / 5.0) |
| GEOS F422 | UG: Geoscience Applications of Remote Sensing (3) | Fall 2018 (4.4 / 5.0) |
| GEOS F497 | UG: Digital Image Processing (3) | Fall 2019 (4.5 / 5.0) |
| GEOS F622 | G: Digital Image Processing in the Geosciences (3) | Fall 2015 (4.2 / 5.0)<br>Fall 2017 (4.1 / 5.0)<br>Fall 2019 (4.6 / 5.0) |

GEOS: Geosciences; NRM: Natural Resource Management; G: Graduate; UG: Undergraduate

*Student course evaluations not available

**A teaching demo from NRM 435 GIS Analysis course (3/5/2024 Lecture Recording)**

**An Introduction to Raster Data**

- Challenge 6 Discussion

- Introduction to Raster Data
- Geospatial tools for raster data processing (Polygon To Raster, Con, Raster Calculator, Clip Raster, Zonal Statistics As Table, Extract Values To Points)

**Massive Open Online Courses on edX.org:**

In summer 2020, I created an open access self-paced 4-week long online course on *Remote Sensing of Wildfires* for the edX platform. **edX.org** is a global nonprofit online education and learning platform founded by Harvard and MIT; their goal of ***increasing access to high-quality education for everyone, everywhere*** aligns well with UAF mission of increasing Open Educational Resources for Alaskans and learners elsewhere. In summer 2021, I created three more open access courses on GIS as part of a GIS Essentials Professional Certificate on the edX platform. So far, we have attracted **42,000+ learners** across the four courses from **190 countries**.

**List of Massive Open Online Courses (on edX.org):**

| ***Course Name*** | ***Duration*** | ***Enrollment*** |
|---|---|---|
| Remote Sensing of Wildfires | 9/29/2020 – present | 3,530 |
| GIS Foundations | 6/15/2021 – present | 19,318 |
| 3D GIS | 6/30/2021 – present | 5,179 |
| GIS Image Analysis | 7/27/2021 – present | 10,687 |

**Graduate Certificate in Geospatial Sciences:** I led a proposal to create a graduate certificate program in Geospatial Sciences. This certificate program became active in fall 2022.

**Guest Lectures:** I have been offering invited guest lectures on a variety of topics (applications of remote sensing and GIS, and permafrost science) to undergraduate and graduate classes since 2012.

**List of invited guest lectures at UAF:**

| ***Lecture Topic*** | ***Course*** | ***Semester*** |
|---|---|---|
| Python GIS Programming | GEOS 458/658: GIS and GPS | Spring 2012 |
| Introduction to Remote Sensing Basic | GEOS 616: Permafrost | Spring 2014 - 2018 |
| Remote Sensing of Permafrost | GEOS 616: Permafrost | Spring 2014 - 2018 |
| Permafrost Modeling in Alaska | IARC Summer School | Summer 2015 - 2017 |
| Thermal Remote Sensing | GE 371: RS for Engineers | Spring 2017 - 2021 |
| Multispectral Transform | GE 371: RS for Engineers | Spring 2018 – 2021 |
| Arctic Permafrost | GeoSTEM pre-service teachers | Fall 2017 |
| Permafrost Modeling in Alaska | NRM 601 Research Methods | Fall 2021, 2023 |
| Permafrost Modeling in Alaska | Surveying course, IIT (ISM) Dhanbad, India | Fall 2021 |

| | | |
|---|---|---|
| Advances in Geospatial Data Science | ESS 492/692 Seminar | Fall 2023, 2025 |

**Student Advising:** I started advising graduate students in fall 2016. I have been part of 8 completed graduate committees, and currently advising 7 graduate students (3 M.S. and 4 Ph.D.).

***Undergraduate Student Advising*:**

| ***Name*** | ***Degree Program*** | ***Duration*** |
|---|---|---|
| Marina Barbosa Santos | Interdisciplinary Undergraduate | 2023 - 2024 |
| Kayla Yanez | Interdisciplinary Undergraduate | 2024 - 2025 |

***Graduate Students Advising as Thesis Committee Chair/Co-Chair:***

| ***Student Name*** | ***Degree*** | ***Status*** |
|---|---|---|
| Barrett Flynn (Co-Chair: Panda) | Ph.D. ESS | Fall 2025 - present |
| Zach Little (Chair: Panda; Co-Chair: Dr. Carlson) | Ph.D. ESS | Fall 2023 - 2024 (Discont.) |
| Joesi Zastrow (Chair: Panda) | M.S. NRE | Inactive |
| Celia Miller (Chair: Panda) | Ph.D. NRE | Fall 2022 - present |
| Sumana Sahoo (Chair: Panda) | Ph.D. NRE | Spring 2022 - present |
| Cristina Ornales (Chair: Dr. Brown, Co-Chair: Panda) | M.S. NRE | **Graduated: 2024** |
| Joshua Monroe (Chair: Panda) | Master NRE | **Graduated: 2023** |
| Anushree Badola (Chair: Panda, Co-Chair: Dr. Bhatt) | Ph.D. Geoscience | **Graduated: 2023** |
| Christopher Smith (Chair: Panda) | M.S. Geography | **Graduated: 2021** |
| Mark Melham (Chair: Dr. Valentine, Co-Chair: Panda) | M.S. NRE | **Graduated: 2019** |
| Cole Payne (Chair: Dr. Prakash, Co-Chair: Panda) | M.S. Geology | **Graduated: 2018** |

***Graduate Students Advising as Thesis Committee Member:***

| ***Student Name*** | ***Degree*** | ***Status*** |
|---|---|---|
| Julia Cheesman (Chair: E. Trochim) | M.S. ESS | Fall 2025 - present |

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|---|---|---|
| Will Samuel (Chair: J. Falke) | M.S. Fisheries | **Graduated: 2023** |
| Michelle Kelley (Chair: Dr. P. Fix) | M.S. NRE | Fall 2021 - 2022 (Discont.) |
| Lora May (Chair: Dr. S. Stuefer) | Interdisc. M.S. | **Graduated: 2024** |
| Natalie Tylor (Chair: Dr. K. Walter Anthony) | Ph.D. Geology | Fall 2019 – present |
| Curtis Bernard (Chair: Dr. E. Nadin) | M.S. Geology | Fall 2018 (Discont.) |
| Maryam A. BuKhader (Chair: Dr. U. Bhatt) | M.S. Atoms. Sc. | **Graduated: 2019** |

***Student Mentoring:***

| ***Student Name*** | ***Degree*** | |
|---|---|---|
| Grace Veenstra | Undergraduate | Summer Field Assistant (2025) |
| Elizabeth Ford | M.S. | Summer Field Assistant (2025) |
| Yesim Goyette | Undergraduate | Summer Field Assistant (2024) |
| Marina Barbosa Santosh | Undergraduate | Summer Field Assistant (2024) |
| Pallavi Prasanna Kumar | M.S. | Research Tech. (2024) |
| Colleen Haan | Undergraduate | Summer Field Assistant (2021) |
| Robert Haan | Undergraduate | Summer Field Assistant (2020) |
| Veronika Doepper | Ph.D. | Summer Field Research (2018) |
| Matthew Whitley | M.S. | Research Assistant (2016) |
| Robert Wagner, Tim Morin, Tyler Hoecker | Undergraduate | IARC Summer School Participants (2015) |

**Postdoc Advising:**

| ***Name*** | ***Research Area*** | ***Duration*** |
|---|---|---|
| Dr. Anushree Badola | Remote Sensing of Boreal Forest Biomass and Tree Mortality | 2023 - 2025 |
| Dr. Christine Waigl | Airborne Hyperspectral Remote Sensing on Boreal Vegetation and Wildfires | 2019 - 2021 |

**SERVICES**

UAF recognizes service in three areas: 1) University Service, 2) Professional Service, and 3) Public Service. Service in all three areas constitutes 10% of my workload.

1) **University Service:** I have served in 13 different university committees since 2016.

- Chair, NRE Graduate Admission Committee, AY2025 – present
- UAF Faculty Contact for MoU between Indian Institute of Technology Roorkee and UAF, AY2025 - present
- Member CNSM Program Review Committee, AY2023-2024
- Member of NRE Assistant Professor Search Committee, AY2023-2024
- Geospatial Science Graduate Certificate, Faculty Coordinator, AY2022 - present
- Member of IANRE Director Search Committee, AY2022-2023
- Member of IANRE IT Recruitment Committee, AY2021-2022
- Member of NRE Graduate Admission Committee, AY2021 – 2025
- Namaste India Student Club Faculty Advisor, AY 2021-2023
- Member of GI Diversity, Equity and Inclusion Committee, AY2020-2021
- FirstGenAK Mentor, AY2020-2021
- Member of UAF Strategic Planning Diversity Committee, AY2018-2019.
- Member of UAF Faculty Senate Research Advisory Committee, AY2018-2020
- GI Faculty Senate (alternate), AY2018-2020
- Member of GI Keith B. Mather Library Advisory Committee (2016 – 2022)

2) **Professional Service:** I have been offering professional service since 2012.

- 2026, judged student research posters during 2026 ESS Research Symposium (2/28/2026)
- Outside examiner, Ph.D. Thesis in Geosciences, Indian Institute of Technology Roorkee, India spring 2026.
- Associate Editor, AmericaView Journal of Earth Observation and Geospatial Applications, AY2024 - present
- Outside examiner, Ph.D. Thesis in Geosciences, Indian Institute of Technology Roorkee, India Fall 2024.
- Chair of AmericaView Education and Outreach Committee, AY2024-2025
- Co-Chair of AmericaView Education and Outreach Committee, AY2023-2024
- Session Co-Chair: International Conference on Permafrost meeting, June 16-20, 2024. Whitehorse, CA.
- Panelist: 2023 Alaska GeoSummit Session: Mapping the needs for Alaska's future geospatial workforce, Oct. 25-26, 2023, Anchorage, Alaska.
- Guest Editor for a special issue in the Geographical Bulletin, Earth from Above: AmericaView, Remote Sensing and Geospatial Technology, 2023.
- Outside examiner, Ph.D. Thesis in Fisheries, UAF, Fall 2023.
- Session Chair: FR3.R14: Satellite Remote Sensing for Water Monitoring at the 2023 IEEE IGARSS meeting held during July 16 - 21, 2023.
- Reviewed applications for the 2022-2023 Fischer Fellowship offered through American Society for Photogrammetry and Remote Sensing (ASPRS) [https://www.asprs.org/]
- 2022-2023 URSA review panel: reviewed UAF Undergraduate Summer Research Academy (URSA) spring and fall 2023 student project award applications.
- Reviewed Alaska NSF EPSCoR Fire and Ice faculty seed grant proposals, Fall 2021 and Fall 2022.
- Session Chair: Advance Geospatial Pathways in Natural Resource Applications at the International Conference on "Geospatial Pathways and Big Data Analytics in Natural Resource Applications and Climate Change" organized by Central University of Jharkhand in association with the Indian Society of Geomatics (ISG), Ranchi Chapter, June 16 - 17, 2022.

- Session Chair: TH1.R4 - Wetlands and Inland Waters I at the 2020 IEEE IGARSS (virtual) meeting held during Sept. 26 - Oct. 2, 2020 (https://igarss2020.org/)
- Reviewed proposals and reports: for U.S. National Park Service, U.S. National Science Foundation, U.S. NASA, Fairbanks North Star Borough.
- Reviewed research articles: for Arctic, Antarctic, and Alpine Research; International Journal of Applied Earth Observation and Geoinformation, International Journal of Remote Sensing, International Journal of Digital Earth, Journal of Geophysical Research: Biogeosciences, Cold Region Science and Technology, Remote Sensing, GIScience, Permafrost and Periglacial Processes, XXIII ISPRS Congress 2016, Arctic Science Summit Week 2016, Ph.D. Thesis from IIT Roorkee (India).
- 2019 Alaska Summer Research Academy (ASRA), UAF: As a co-instructor I taught a middle school ASRA module on "*The burning forests of Alaska*". Students learned about remote sensing basics and its application in forest fire research. It included lecture, hands-on lab exercises, and field visit to a burned site within Caribou-Poker Creeks Research Watershed. Web link: https://www.uaf.edu/asra/the_burning_forests_of_alaska.php
- 2019 Upward Bound College Bound program, UAF: Offered two lectures on remote sensing application to middle and high-school student participants of Upward Bound College Bound summer program. It included hands-on lab exercise on application of remote sensing on coastal erosion research, and lab on Google MyMap application.
- Reviewed the Cryosphere chapter of the Fairbanks North Star Borough's current Hazard Mitigation Plan in spring 2019.
- Reviewed applications for the Next Generation of Polar Researchers Leadership Symposium in spring 2019. https://dornsife.usc.edu/conferences/2019-nsf-ngpr3/
- Twice (summer 2018 and spring 2019) served as reviewer for the NSF Navigating the New Arctic proposal review panel.
- As part of an Informal STEM Learning (ISL) project funded through the NSF titled "*Hot Times in Cold Places: The Hidden World of Permafrost*", I contributed to the development of a 2000 square feet **traveling exhibit** "Under the Arctic: Digging into Permafrost" on permafrost and climate change in Alaska. The traveling exhibit was developed by Oregon Museum of Science and Industry (OMSI). www.permafrosttunnel.org
- Served as Editor of Changing Ice: A Newsletter of Cryosphere Research in Alaska in 2105-2016.
- Judged student presentations for Outstanding Student Presentation Award at the 2014 and 2015 American Geophysical Union Fall Meeting held during December in San Francisco, CA.
- Led a site visit for 2012 Friends of Pleistocene field excursion on geology and geo-hazards along Alaska Highway and Glenn Highway.

3) **Public Service:** I have been doing public service as a UAF employee since 2012.

- 12/4/2025, Spoke to Ms. Z's 6th grade class at the Boreal Sun chartered school about Monsoon in India.
- Outreach on satellite earth observation and benefit to society, and remote sensing of forest health at the following events:
    - 2019, 2024 - 2026 CNSM Science Potpourri
    - 2023 - 2025 Pearl-Creek Elementary School Science Night
    - 2022 - 2026 Arctic Research Open House

    - 2022 ArcticFest held during Sept. 2 -3, 2022
- 2023 - 2025: Manned IANRE booth at the Tanana Valley State Fair
- I have judged K-12 science fair projects at Interior Alaska District Science Fair (2015, 2016, 2017, 2018, 2022, 2023, 2025), Catholic Schools of Fairbanks Science Fair (2016 and 2018), and HomeSchool Science Fair (2016, 2017, and 2018).
- Presented lectures on Permafrost to volunteers of Oregon Museum of Science and Industry (Nov. 2017), to middle school students of Alaska Summer Research Academy Permafrost Module class (July 2017, 2018), and to Geology 1064 class of Oklahoma City Community College (Oct. 2015).
- Since 2014 I have conducted outreach on permafrost and climate change science at schools and **21 Alaskan communities** (Tok, Northway, Mentasta, Anderson, Shishmaref, Delta Junction, Dot Lake, Healy, Allakaket, Nikolai, Nenana, Fairbanks, Ruby, Tanana, Rampart, Utquiagvik, Kotzebue, Stevens Village, Beaver, Fort Yukon, and Bethel).
- I am a "Competent Person" authorized to give a science tour of the CRREL Permafrost Research Tunnel. In recent years I have taken Elementary school children to Government officials and Oil companies executives through the tunnel. (http://permafrosttunnel.crrel.usace.army.mil/). Since 2014 I have given more than two dozen science tours of the research tunnel.
- Outreach at CRREL Permafrost Tunnel Research Facility: Contributed to hugely successful Permafrost Tunnel Open House during August 18-19, 2012. This was a unique opportunity for the general public and kids to get a tour of the tunnel and learn about climate change, permafrost environment, and last Ice Age fauna and flora. About 2,000 people visited the tunnel.
- As part of the Fairbanks Team in Training, I raised $4,352 for Leukemia & Lymphoma Society (LLS) during summer 2015 – 2017.

**RESEARCH**

My research career started with a Research Fellow position in the lab of Dr. A. K. Saraf at Indian Institute of Technology, Roorkee (India) in 2005. There, I processed and analyzed AVHRR, MODIS and Landsat LST data to investigate earthquake precursors in Southeast Asia. It resulted in several peer-reviewed articles (one as a lead author in International Journal of Remote Sensing), and several conference proceedings and presentations.

In 2007, I moved to Fairbanks, Alaska to pursue a Ph.D. in Geology from University of Alaska Fairbanks. Under the supervision of Dr. A. Prakash, I studied the distribution and temperature dynamics of perennially frozen ground also known as permafrost. My research included field data collection, mapping and modeling of near-surface permafrost distribution along a 100 x 12-mile stretch of Alaska Highway between Delta Junction and Tok using multi-source remote sensing and field datasets. Also, modeled permafrost temperature dynamics using the GIPL 2.0 numerical model. It resulted in several peer-reviewed articles (two as the lead author), and several conference proceedings and presentations.

After completing my Ph.D. in 2011, I continued at UAF as a Postdoctoral Fellow in the Snow, Ice and Permafrost research group at the Geophysical Institute. My postdoctoral research included monitoring of permafrost temperature from a series of deep boreholes along the 800-mile-long north-south transect from Prudhoe Bay to Gulkana, Alaska, and regional scale modeling of permafrost temperature dynamics. I produced the first ever medium-resolution (30-m) permafrost distribution and temperature maps for the seven national park units of Alaska and coarse-resolution (771-m) maps for the North Slope region of Alaska. Also, during this period I landed my first NSF research grant as a co-PI for a project focused on community capacity building for permafrost and climate monitoring in rural Alaska.

In 2016, I was promoted to Research Assistant Professor position and expanded my research beyond permafrost to river erosion, forest vegetation, and wildfires.

- Application of hyperspectral remote sensing for vegetation/fire fuel mapping, active fire characterization, and forest fire severity assessment
- Applications of GIScience and geophysics to study permafrost-affected landscape changes and related geo-hazards, and assessing their societal and environmental impacts
- Applications of GIScience and remote sensing on river processes (e.g. bank erosion, channel shift, and bathymetry)

In 2020, I joined the Department of Natural Resources and Environment (NRE), and Institute of Agriculture, Natural Resources and Extension (IANRE) as a tenure-track Associate Professor and received tenure in July 2023. In my current position, I am developing new research programs on remote sensing of forest health and natural resource management.

**PROJECTS**

At UAF, so far I have contributed to 13 completed projects in various capacities and currently I have 4 active projects. Overall, I have contributed to 17 projects totaling ~$29 million (as PI/ Co-PI: $1.7 million).

***ACTIVE PROJECTS***

1. *FireAID: An Undergraduate Research Training Program to Develop Technologies to Fight Wildland Fire with Artificial Intelligence and Deep-Learning in Alaska*. DOE, Role: Co-PI, Award period: 2025 - 2028, Total funding: $2,250,000
2. *Remote sensing based mapping, monitoring, and modeling of tree mortality in Alaska's boreal forest.* USDA NIFA. Role: Co-PI, Award period: 2024 - 2026, Total funding: $100,000
3. *Remote sensing of forest health, disturbance, and change, Alaska.* Funded by USDA NIFA. Role: PI, Award period: 2021 – 2026, Total funding: $786,887
4. *StateView program development and operations for the state of Alaska*. Funded by the U.S. Geological Surveys. Role: PI, Award period: 2023 – 2028, Total funding: $127,500

***COMPLETED PROJECTS***

1. *Fire and Ice: Navigating variability in boreal wildfire regimes and subarctic coastal ecosystems.* Funded by the NSF. Role: Co-I, Award period: 2018 – 2024, Total funding: $20,000,000
2. *StateView program development and operations for the state of Alaska*. Funded by the U.S. Geological Surveys. Role: PI, Award period: 2018 – 2023, Total funding: $117,500
3. *Improving remote sensing based boreal wildfire burn severity and risk assessment, interior Alaska*. Funded by Alaska Space Grant Program. Role: Faculty Advisor, 2020 – 2021, Total funding: $42,610 (includes 100% match)
4. *Community based permafrost and climate monitoring in rural Alaska*. Funded by the NSF. Role: Co-PI, 2015 – 2019, Total funding: $460,778
5. *Hot Times in Cold Places: The Hidden World of Permafrost*. Funded by the NSF. Role: Co-I, Award period: 2014 – 2019, Total funding: $1,755,019
6. *Remote sensing of Colville River navigability to aid subsistence travel, Nuiqsut, Alaska.* Funded by Alaska Space Grant Program. Role: Faculty Advisor, 2017 – 2018. Total funding: $37,070
7. *Changing Colville River environment and its impact on river navigation and subsistence travel.* Role: Faculty advisor, 2017 – 2018. Total funding: $9,964
8. *Next-Generation Ecosystem Experiments (NGEE) Arctic*. Funded by the U. S. Dept. of Energy. Role: Co-I, Award period: 2014 – 2019, Total funding: $2,970,000

9. *Sixty-five years of Colville River dynamics and its impact on the present river navigability near Nuiqsut, North Slope of Alaska*. Funded by the Alaska EPSCoR Seed Grant 2016. Role: PI, Total funding: $72,441
10. *Human perceptions and consequences of a changing permafrost landscape near Nuiqsut, Alaska*. Funded by the Alaska EPSCoR Native Engagement Grant 2016. Role: Co-PI, Total funding: $29,900
11. *Use of AIEM permafrost module output to assess the permafrost changes in the 21st century.* Funded by the USGS Alaska CSC. Role: Postdoc, Total funding: $189,198
12. *Permafrost modeling in Alaskan National Park Lands*. Funded by the National Park Service (NPS). Role: Postdoc, Total funding: $371,097
13. *AON: Thermal State of Permafrost (TSP) in North America and Northern Eurasia: The US contribution to the International Network of Permafrost Observatories (INPO)*. Funded by the NSF. Role: Postdoc, Total funding: $1,859,861
14. *Modeling of permafrost dynamics along Alaska Highway corridor, Interior Alaska*. Funded by Center for Global Change and Arctic System Research, International Arctic Research Center (University of Alaska Fairbanks). Role: PI, Total funding: $8000

**PROPOSALS *IN REVIEW***

**PROPOSALS *IN PREPARATION***

1. *Growing Alaska's hyperspectral remote sensing capacity for mapping natural resources and monitoring geo-hazards*. 2025 NSF Arctic Natural Science Program.
2. *Training the next generation of Natural Resource specialists and scientists for Alaska*. 2025 NSF Research Experience for Undergraduates (REU), Directorate of Geosciences.

***DECLINED PROPOSALS*** *(not awarded but have potential to be revised and resubmitted)*

1. *Studying North Alaskan permafrost to assess hazards, study microbiome, and analyze paleotemperature*. DOE, Role: Co-PI, Award period: 2024 - 2027. Total funding: $150,000
2. *Multi-Sensor Remote Sensing Approach to Advance Early Detection of Insect Pest Damage in Alaska's Forest.* NASA FINNEST. Role: PI (Faculty Mentor), Award period: 2025 - 2026, Total funding: $50,000
3. *Fusing process-based models with airborne remote sensing to understand Alaskan boreal forest ecology and sensitivity to climate change*. Submitted to 2021 NASA ABoVE Phase 3 Terrestrial Ecology program, in collaboration with University of California Santa Barbara. Role: Institutional PI, Total budget: $174,644.
4. (Sub)arctic shrub biomass estimation from L- and S-band SAR. Submitted to 2019 NASA program *Utilization of airborne L- and S- band synthetic aperture radar imagery over North America – Joint NASA and ISRO airborne campaign*. Role: Co-I, Total budget: $149,987.
5. *Freshwater ice thickness mapping using remote sensing to aid safe winter travel in interior Alaska*. Alaska NASA EPSCoR 2017. Role: PI. $ 30,100
6. Collaborative Research: *Sustained observations of socio-ecological impacts and biophysical drivers of landscape change in boreal-tundra regions of Alaska*. NASA Citizen Science for Earth Systems Program 2016. Role: PI. $700,000.
7. Collaborative Research: *A study of thermokarst dynamics using InSAR, field measurements, and modeling.* 2015 NSF Arctic Research Opportunity (Arctic Natural Sciences Program). Role: PI. $139,086.

8. Collaborative Research: *Potential impacts of surface and subsurface water dynamics on permafrost thermal dynamics and associated soil carbon feedbacks*. 2015 NSF Arctic Research Opportunity (Arctic Natural Sciences Program). Role: Co-PI. $459,739.
9. Collaborative Research: *Vulnerable hot spots for the Permafrost Carbon Feedback*. 2015 NASA Terrestrial Ecology Program. Role: Co-I. $ 602,080.00
10. Collaborative Research: *Using InSAR, field measurements, and models to study thermokarst dynamics*. 2014 NSF Arctic Research Opportunity (Arctic Natural Sciences Program). Role: PI. $140,000.
11. Outreach: *Art inspired by the UAF Geophysical Institute*. 2014 University of Alaska Fairbanks People's Endowment Grant. Role: Co-PI. $5000.
12. Collaborative Research: *Ground Ice, Methane, and the Permafrost Carbon Feedback*. 2013 NASA Research Opportunity in Space and Earth Sciences (ROSES-2013, A.5 Carbon Cycle Science). Role: PI. $155,966.
13. Snow, Ice, and Permafrost Traveling Exhibit. 2013 UAF Peoples Endowment Grant 2013. Role: Co-PI. $5000

**LIST OF PUBLICATIONS AND PRESENTATIONS** (first author advisee names are underlined)

Google Scholar Citation Page (Total citation: 1,783; h-index: 18; i10-index: 22)

***BOOKS***

1. Prakash A., Kuenzer C., **Panda, S.K.**, Badola A.,Waigl, C. 2024. Remote sensing based mapping and monitoring of coal fires in Remote Sensing Handbook, Second Edition, Vol. 6: Drought, Disaster, Pollution and Urban Planning, Edited by Prasad Thenkabail, Routledge, Taylor & Francis Group. ISBN 9781032580586.
2. Bar, Somnath, B. R. Parida, **S.K. Panda**. Changing forest fire regime with relation to climate conditions over western and eastern Himalaya, India in Handbook of Himalayan Ecosystems and Sustainability, Vol. 1: Spatio-Temporal Monitoring of Forests and Climate, Edited by B.R. Parida, CRC Press, Taylor & Francis Group. ISBN 9781032203140.
3. Cysewski, S., M. Sturm, **S.K. Panda**, and M. Cysewski. Nuiqsut on the Colville: Land of Sky, Tundra, and Water. A photo journal documenting Nuiqsut and its changing landscape for the people who live there.

***PEER-REVIEWED JOURNALS*** [Citations]

1. **Panda, S.K.**, and Badola, A., 2026. A remote sensing perspective on Alaska's changing forest ecosystem: a review. *Rem. Sen. of Env.* (in preparation)
2. **Panda, S.K.**, and Das, A.K., 2026. Remote sensing of wildfires in Alaska: Opportunities and Challenges, *Forests* (in preparation)
3. Enguehard, L. Kruse, S., Hänsch, R., Herzschuh, U., Panda, S., Heim, B. (2025) Exploring Harmonized Landsat-Sentinel-2 satellite data for predicting boreal forest structure from UAV-LiDAR data in Northwestern America. Science of Remote Sensing. https://doi.org/10.1016/j.srs.2026.100403
4. Soil moisture and active layer thickness across Alaska USA and Northwestern Canada. Scientific Data (in review, submitted on Oct. 11th)
5. Badola, A., **Panda, S.K.**, Ford, E., Juday, G.P., Gens, R. 2026. An integrated remote sensing approach for improved forest biomass quantification in boreal Alaska. *GIScience and Remote Sensing.* (in review)

6. Sahoo, S., **Panda, S.K.**, et al. 2026. A novel algorithm to detect Aspen leaf-miner infestation and severity in interior Alaska. *Scientific Report Special Issue.* (in review)

7. Enguehard, L. et al. (2025) Investigating Boreal Forest Successional Stages in Alaska and Northwest Canada Using UAV-LiDAR and RGB and a Community Detection Network. *Remote Sensing in Ecology and Conservation.* http://doi.org/10.1002/rse2.70029

8. Schore, A.I.G., Wall, W.A., Panda, S.K., et al. (2025) Plant functional types improve satellite-derived burn severity assessments in interior Alaska. *Environmental Research: Ecology* (rejected)

9. Samuel, W.T., K.D. Tape, A.C. Seitz, **S.K. Panda,** J.A. Falke, 2026. When beavers get burned, do fish get fried? The role of beavers in mediating wildfire effects on Arctic Grayling in boreal Alaska. *Ecosphere*. (in review)

10. Sahoo, S., Juday, G.P., **Panda, S.K.**, et al. 2025. Interplay of topography, fire history, and climate on Interior Alaska boreal vegetation dynamics in the 21st century: A Landsat time-series analysis. *Forests.* 16(5), 777. https://doi.org/10.3390/f16050777

11. Waigl, C., Badola, A., **Panda, S.K.**, et al. 2024. Boreal forests at tree-scale and stand-scale: Exploring the benefits of in-house aerial VNIR/SWIR imaging spectroscopy for Alaska wildfire research. *Remote Sensing.* (in preparation)

12. Waigl, C., Schmidt, J., Greaves, Brix, Stuefer, M., **Panda, S.K.**, Berman, M., Bhatt, U. 2024, The 2019 McKinley fire in south-central Alaska: burn severity and fire effects from high-resolution aerial imaging spectroscopy. *International Journal of Applied Earth Observations and Geoinformation* (in preparation).

13. Badola, A., **Panda, S.K.**, Thompson, D., Roberts, D.A., Waigl, C., and Bhatt, U.S. 2023, Estimation and validation of sub-pixel needleleaf cover fraction in the boreal forest of Alaska to aid wildfire management. *Remote Sensing, 15(10)*: 2484*, https://doi.org/10.3390/rs15102484*.

14. Bar, S., Parida, B.R., Pandey, A.C., Shankar, B.U., Kumar, P., **Panda, S.K.**, and Behera, M.D. 2023. Modeling and prediction of fire occurrences along an elevational gradient in Western Himalayas. *Applied Geography,* V. 151 (102867). https://doi.org/10.1016/j.apgeog.2022.102867 [1]

15. Badola, A., **Panda, S.K.**, Roberts, D.A., Waigl, C., Jandt, R.R., Bhatt, U.S. 2022. A novel method to simulate AVIRIS-NG hyperspectral image from Sentinel-2 image for improved vegetation/wildfire fuel mapping, boreal Alaska. *International Journal of Applied Earth Observation and Geoinformation,*. V. 112 (102891)*. https://doi.org/10.1016/j.jag.2022.102891* [9]

16. Smith, C.W., **Panda, S.K.**, Bhatt, U.S., Meyer, F.J., Badola, A., Hrobak, J.L. 2021. Assessing wildfire burn severity and its relationships with environmental factors: a case study in interior Alaska boreal forest. *Remote Sensing, 13(10): 1966*. https://doi.org/10.3390/rs13101966 [10]

17. Badola, A., **Panda, S.K.**, Roberts, D.A., Waigl, C., Bhatt, U.S., Smith, C.W., Jandt, R.R. 2021. Hyperspectral data simulation (Sentinel-2 to AVIRIS-NG) for improved wildfire fuel mapping, boreal Alaska. *Remote Sensing*, *13(9): 1693*. https://doi.org/10.3390/rs13091693 [15]

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19. Doepper, V., **Panda, S.K.**, Waigl, C., Braun, M., and Feilhaur, H. 2021. Using floristic gradient mapping to assess seasonal thaw depth in interior Alaska. *Applied Vegetation Science*. https://onlinelibrary.wiley.com/doi/10.1111/avsc.12561 [6]

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Wu, T., Yamkin, J., Zou, D., 2019, Northern Hemisphere permafrost map based on TTOP modeling for 2000-2016 at 1 $km^2$ scale. *Earth Sciences Reviews, 193: 299-316*. [685]

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23. **Panda, S.K.**, Romanovsky, V.E., Marchenko, S.S., and Swanson, D.K., 2017, The fate of permafrost. *Alaska Park Science,* 16(1): 39-46.
24. Jafarov, E.E., Parsekian, A.D., Schaefer, K., Liu, L., Chen, A., **Panda, S.K.**, and Zhang, T., 2017. Estimating active layer thickness and volumetric water content from ground penetrating radar measurements in Barrow, Alaska. *Geoscience Data Journal*. DOI: 10.1002/gdj3.49 [27]
25. Nicolsky, D.J., Romanovsky, V.E., **Panda, S.K.**, Marchenko, S.M., and Muskett, R.R., 2017, Applicability of the ecosystem type approach to model permafrost dynamics across the Alaska North Slope. *Journal of Geophysical Research*, 122(1): 50-75. [89]
26. O'Donnell, J.A., Aiken, G.R., Swanson, D.K., **Panda, S.K.**, Butler, K.D., and Baltensperger, A., 2016, DOM composition of Arctic rivers: linking permafrost and parent material to riverine carbon. *Global Biogeochemical Cycles*, 30(2): 1811-1826. [79]
27. Chen, A., Parsekian, A., Schaefer, K., Jafarov, E., **Panda, S.K.**, Liu, L., Zhang, T., and Zebker, H., 2016, Ground-penetrating radar measurements of active-layer thickness at landscape scale on the North Slope of Alaska. *Geophysics*, 81(2):h1-h11. http://library.seg.org/doi/abs/10.1190/geo2015-0124.1 [27]
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32. **Panda, S.K.,** Prakash, A., Solie, D.N., Romanovsky, V.E. and Jorgenson, M.T., 2010**,** Remote sensing and field-based mapping of permafrost distribution along the Alaska Highway corridor, Interior Alaska. *Permafrost and Periglacial Processes*, 21(3), 271-281. [56]
33. Saraf, A.K., Rawat, V., Banerjee, P., Choudhury, S., **Panda, S.K.**, Dasgupta, S. and Das, J.D., 2008, Satellite detection of earthquake thermal infrared precursors in Iran. *Natural Hazards*, 47, 119-135. [132]
34. Saraf, A.K., Choudhury, S., **Panda, S.K.**, Dasgupta, S. and Rawat, V., 2007, Does a major earthquake precede a thermal anomaly. *International Journal of Geoinformatics*, 3(3). [8]
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36. Das, J.D., Saraf, A.K., and **Panda, S.K.**, 2007, Satellite data on rapid analysis of Kashmir earthquake (October 2005) triggered landslide pattern and river water turbidity in and around epicentral region. *International Journal of Remote Sensing*, 28(8), 1835-1842. [23]
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38. Choudhury, S., Saraf, A.K., and **Panda, S.K.**, 2006, SSM/I Application in Studies of Thermal Anomalies Associated with Earthquakes. *International Journal of Geoinformatics*, 2(3), 9-16.
39. Choudhury, S., Dasgupta, S., Saraf, A.K., and **Panda, S.K.**, 2006, Remote sensing observations of pre-earthquake thermal anomalies in Iran. *International Journal of Remote Sensing,* 27(20), 4381-4396. [119]

***REPORTS***

1. **Panda, S.K.**, Whitley, M.A., 2017, Sixty-five years of Colville River dynamics and its impact on present river navigability near Nuiqsut, North Slope of Alaska. *Alaska EPSCoR Project Report, 19 p.*
2. **Panda, S.K.**, Romanovsky, V.E., Marchenko, S.S., 2016, High-resolution permafrost modeling in the Arctic Network National Parks, Preserves and Monuments. *Natural Resource Technical Report* NPS/ARCN/NRTR-2016/1366. National Park Service, Fort Collins, Colorado. https://irma.nps.gov/Datastore/Reference/Profile/2237720 [17]
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4. **Panda, S.K.**, Marchenko, S.S., Romanovsky, V.E., 2014, High-resolution permafrost modeling in Wrangell-St. Elias National Park and Preserve. *Natural Resource Technical Report* NPS/CAKN/NRTR-2014/861. National Park Service, Fort Collins, Colorado. https://irma.nps.gov/Datastore/Reference/Profile/2209250 [4]

***PUBLISHED DATA AND PRODUCTS***

1. Sahoo, S., Lewis, R., **Panda, S.K.**, Hutten, K., Brown, D., Juday, G., & Das, A. K. (2026). Forest Trend Mapper v1.0.0 (v1.0.0). Zenodo. https://doi.org/10.5281/zenodo.19865390
2. Badola A., Kumar, P.P., **Panda, S.K.** 2024. Hyperspectral image simulation application for Alaska. https://zenodo.org/records/13973468
3. Badola A., **Panda, S.K.** 2023. AVIRIS-NG data simulation and classification for generating vegetation maps for boreal Alaska. https://doi.org/10.5281/zenodo.8277600
4. Schaefer, K., L.K. Clayton, M.J. Battaglia, L.L. Bourgeau-Chavez, R.H. Chen, A.C. Chen, J. Chen, K. Bakian-Dogaheh, T.A. Douglas, S.E. Grelick, G. Iwahana, E. Jafarov, L. Liu, S. Ludwig, R.J. Michaelides, M. Moghaddam, S. Natali, **S.K. Panda**, A.D. Parsekian, A.V. Rocha, S.R. Schaefer, T.D. Sullivan, A. Tabatabaeenejad, K. Wang, C.J. Wilson, H.A. Zebker, T. Zhang, and Y. Zhao. 2021. ABoVE: Soil Moisture and Active Layer Thickness in Alaska and NWT, Canada, 2008-2020. ORNL DAAC, Oak Ridge, Tennessee, USA. https://doi.org/10.3334/ORNLDAAC/1903 [5]
5. Nicolsky, D.J., Romanovsky, V.E., **Panda, S.K.**, Marchenko, S.M., and Muskett, R.R., 2018, Simulated Permafrost Dynamics and Thaw Subsidence across the Alaskan North Slope Region in the 20th and 21st Centuries. USGS CSC Archive.
6. Jafarov, E., A. Parsekian, K. Schaefer, L. Liu, A. Chen, **S.K. Panda**, and T. Zhang. 2016. Pre-ABoVE: Active Layer Thickness and Soil Water Content, Barrow, Alaska, 2013. ORNL DAAC, Oak Ridge, Tennessee, USA. http://dx.doi.org/10.3334/ORNLDAAC/1355 [2]

7. Chen, A., A. Parsekian, K. Schaefer, E. Jafarov, **S.K. Panda**, L. Liu, T. Zhang, and H.A. Zebker. 2015. Pre-ABoVE: Ground-penetrating Radar Measurements of ALT on the Alaska North Slope. Available on-line [http://daac.ornl.gov] from Oak Ridge National Laboratory Distributed Active Archive Center, Oak Ridge, Tennessee, USA. http://dx.doi.org/10.3334/ORNLDAAC/1265 [3]

8. Liu, L., K. Schaefer, A. Chen, A. Gusmeroli, E. Jafarov, **S.K. Panda**, A. Parsekian, T. Schaefer, H. A. Zebker, T. Zhang. 2015. Pre-ABoVE: Remotely Sensed Active Layer Thickness, Barrow, Alaska, 2006-2011. Data set. Available on-line [http://daac.ornl.gov] from Oak Ridge National Laboratory Distributed Active Archive Center, Oak Ridge, Tennessee, USA. http://dx.doi.org/10.3334/ORNLDAAC/1266 [1]

9. Liu, L., K. Schaefer, A. Chen, A. Gusmeroli, E. Jafarov, **S.K. Panda**, A. Parsekian, T. Schaefer, H. A. Zebker, T. Zhang. 2015. Pre-ABoVE: Remotely Sensed Active Layer Thickness, Prudhoe Bay, Alaska, 1992-2000. Data set. Available on-line [http://daac.ornl.gov] from Oak Ridge National Laboratory Distributed Active Archive Center, Oak Ridge, Tennessee, USA. http://dx.doi.org/10.3334/ORNLDAAC/1267

***IN CONFERENCE/ SYMPOSIUM PROCEEDINGS***

1. Badola, A., Juday, G.P., Panda, S.K., and Kruse, S. 2024, Tracking spatial and temporal variation in forest canopy conditions in the Boreal forest of Alaska using Lidar data. In the Proceedings of *2024 IGARSS* held during July 7 – 12 in Athens, Greece.

2. Greer, P., Ulinski, A., Khan, S.D., et al. 2023, Understanding the change in permafrost by imaging the CRREL Permafrost tunnel, Fairbanks, Alaska. SEG International Exposition and Annual Meeting.

3. Badola, A., Panda, S.K., Juday, G.P. 2023, Effect of burn severity on needleleaf recovery: A case study of 1983 Rosie Creek fire, Interior Alaska. In the Proceedings of *2023 IGARSS* held during July 16 – 21 in Pasadena, California.

4. Waigl, C.F., Stuefer, M., Badola, A., Smith, C., **Panda, S.K.**, and Bhatt, U.S. 2022, Boreal forests at tree-scale and stand-scale: Exploring the benefits of in-house aerial VNIR/SWIR imaging spectroscopy for Alaska wildfire research. In the proceedings of the *16th International Circumpolar Remote Sensing Symposium* held during May 16 – 20, 2022 at University of Alaska Fairbanks Campus, Fairbanks, Alaska, USA.

5. Badola .A., **Panda, S.K.**, Roberts, D., Bhatt, U.S., Waigl, C., Jantd, R. 2022. Addressing the hyperspectral data needs for the Arctic environment through image simulation. In the proceedings of the *16th International Circumpolar Remote Sensing Symposium* held during May 16 – 20, 2022 at University of Alaska Fairbanks Campus, Fairbanks, Alaska, USA.

6. **Panda, S.K.**, Payne, C**.**, Smith, C., Prakash, A., Brinkman, T. 2020, Mapping of shallow water sites to aid navigation on the Colville River, North Slope of Alaska. *2020 IGARSS* held (online) during Sept. 28 – Oct. 2, 2020. https://ieeexplore.ieee.org/document/9323229 [1]

7. Smith, C., **Panda, S.K.**, Bhatt, U.S., and Meyer, F.J., Haan, R.W. 2020, Improved vegetation and wildfire fuel type mapping using NASA-NG AVIRIS Hyperspectral data, Interior Alaska. *2020 IGARSS* held (online) during Sept. 28 – Oct. 2, 2020. https://ieeexplore.ieee.org/document/9324136 [1]

8. **Panda, S.K.**, Kholodov, A., Dubay, C., and Hansen, T., 2016, Engaging rural communities in permafrost and climate monitoring in the Upper Kuskokwim region, interior Alaska, Arctic Observing Summit, held Mar.15-18, 2016 in Fairbanks, Alaska. (https://assw2016.org/about/aos)

9. **Panda, S.K.**, Prakash, A. and Solie, D.N., 2008a, Remote sensing-based study of vegetation distribution and its relation to permafrost in and around George Lake area, central Alaska. In the proceedings of *Ninth International Conference on Permafrost*, held June 29-July 3 at University of Alaska Fairbanks, Fairbanks, Alaska, Vol. II, 1357-1362. [6]

10. Saraf, A.K., Rawat, V., Choudhury, S., Banerjee, P., Dasgupta, S., **Panda, S.K.**, and Das, J.D., 2008, Is satellite detection of earthquake precursor possible?, In the proceedings of *European Seismology Commission 31 General Assembly (ESC 2008)*, held Sept. 7-12 in Crete, Greece.
11. Saraf, A.K., Choudhury, S., **Panda, S.K**., and Dasgupta, S., 2007, Does a major earthquake precede a thermal anomaly? In the proceedings of *1st International Workshop & Conference of Geoinformation Technology for Natural Disaster Management and Rehabilitation*, held May 8-9 in Esfahan, Iran, organized by Asian Institute of Technology, Bangkok & Association of Geoinformation Technology, Bangkok, pp. 9.
12. Saraf, A.K., Choudhury, S., **Panda, S.K**., and Dasgupta, S., 2007, Remote Sensing Observations of Pre-earthquake Thermal Anomalies in Iran. In the proceedings of *5th Conference on Seismology and Earthquake Engineering (SEE 5)*, held May 14-16 at International Institute of Earthquake Engineering and Seismology, Tehran, Iran.
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14. Das, J.D., Saraf, A.K. and **Panda, S.K**., 2006, Kashmir Earthquake of 2005 generated Landslides and Water Turbidity as detected by Satellite Data and the Tectonic setup. In the proceedings of *13th Symposium on Earthquake Engineering (13 SEE-06)*, held December 19-21 at Department of Earthquake Engineering, Indian Institute of Technology Roorkee, India, Vol. I, 335-341.
15. Das, J.D., Choudhury, S., Saraf, A.K. and **Panda, S.K**., 2006, Earthquake Hazard Perception in Seismically Active Northeastern Part of India being Affected by the Analogous Sumatran Tectonic Setting. In the proceedings of *13th Symposium on Earthquake Engineering (13 SEE-06)*, held December 19-21 at Department of Earthquake Engineering, Indian Institute of Technology Roorkee, Roorkee, Vol. I, 98-106.
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17. **Panda, S.K.** and Saraf, A.K., 2005, Application of Remote sensing & GIS in post-tsunami damage assessment. In the proceedings of *Symposium on Seismic Hazard Analysis and Microzonation*, held September 23-24 at Dept. of Earthquake Engineering, Indian Institute of Technology Roorkee, Roorkee, India (Roorkee: Paramount Offset Printers), 435-442. [2]

***PRESENTATIONS IN CONFERENCES/ SYMPOSIUMS/ WORKSHOPS*** **(P: Poster; T: Talk)**

1. Sahoo, S. and Panda, S. Forest Trend Mapper: A Cloud Computing Tool for Tracking Landscape Change in Alaska's Boreal Forests, IALE-North America 2026 Meeting, April 26-20, Athens, GA.
2. Sahoo, S. and Panda, S., Aspen leafminer detection in Interior Alaska from high-resolution satellite data using deep learning, ForestSAT 2026 Conference, May 4-8, Gainesville, FL.
3. Das et al., Igniting Alaska's AI Future: Developing Alaska's AI Workforce Through Wildfire Analysis, Trillion Parameters Consortium 2026 All Hands Conference, May 31-June 3, Baltimore, MD.
4. Heim, B., and Lea, E et al. AI-vergreens - a New Multi-level Labelled Multi-temporal of Harmonized Landsat and Sentinel-2 (HLS) based Training Dataset Optimized for Northern Circumboreal Forest, 2025 AGU Fall Meeting, Dec. 15-19, New Orleans, Louisiana [Abstract, P]
5. Ozturk, D.S. et al., Arctic-Focused Geophysical and Interdisciplinary Learning Through AlaskaX, 2025 AGU Fall Meeting, Dec. 15-19, New Orleans, Louisiana. [Abstract, P]

6. Kholodov, A.L. et al., Permafrost dynamics in the region of upper Kuskokwim and its impact on the process of pedogenesis. 2025 AGU Fall Meeting, Dec. 15-19, New Orleans, Louisiana. [Abstract, P]
7. Sahoo, S., **Panda, S.K.** 2025. Comparative Analysis of Aspen Leaf Miner Detection and Severity Assessment in interior Alaska Using Multi-Resolution Remote Sensing, presented at the 2025 Alaska Geosummit, Apr. , 2025.
8. Sahoo, S., Prasannakumar, P., **Panda, S.K.** 2025. Alaska's changing forest - Intro to forest trend mapper, will be presented at the 2025 Alaska Forum on Environment, Feb. 3-7 , 2025.
9. Sahoo, S., Badola, A., Hutten, K., **Panda, S.K.** 2024. Remote sensing of forest health: an open forum for ideas and insights, presented at 2024 Alaska Forum on Environment Virtual Technical Session on Nov. 13th, 2024. Recording
10. Lea. E., Heim B., Juday, G.P., **Panda, S.K.** et al. 2024. Characterizing boreal forest successional stages and disturbance impacts on forest structure using UAV LiDAR and multispectral data across Alaska and Canada, 2024 AGU Fall Meeting, Dec. 9-13, Washington D.C. [Abstract, P]
11. Badola .A., **Panda, S.K.,** Juday, G.P., Gens, R. 2024. Time-series analysis of forest mortality trends using satellite imagery and ground-based data in boreal forests, interior Alaska, 2024 AGU Fall Meeting, Dec. 9-13, Washington D.C. [Abstract, P]
12. Badola, A., **Panda, S.K.**, and Juday, G.P. 2024. Integrating UAVSAR and LiDAR Data for Above-Ground Biomass Estimation in White Spruce Forests, Boreal Alaska at 2024 American Association of Geographers Annual Meeting, Apr. 16 - 20, Honolulu.
13. Sahoo, S., Meyer, F.J., **Panda, S.K.**, 2023. Monitoring Forest Disturbances in Alaska using Polarimetric synthetic aperture radar (PolSAR) at 2023 AGU Fall Meeting, Dec. 9-13, San Francisco.
14. Samuel, W. T., J. A. Falke, K. D. Tape, **S. K. Panda**, A. C. Seitz. 2023. When beavers get burned, do fish get fried? Assessing beaver effects on fish in a fire-dominated ecosystem using a combination of traditional fish sampling and eDNA. American Fisheries Society Student Subunit Annual Research Symposium, Fairbanks/Juneau, AK. February 24.
15. Samuel, W. T., J. A. Falke, K. D. Tape, **S. K. Panda**, A. C. Seitz. 2023. When beavers get burned, do fish get fried? Assessing beaver effects on fish in a fire-dominated ecosystem using a combination of traditional fish sampling and eDNA. Alaska Chapter American Fisheries Society Annual Meeting, Fairbanks, AK. March 27-31.
16. Badola .A., Thompson, D., Roberts, D., **Panda, S.K.,** Waigl, C., Bhatt, U.S. Jantd, R. 2022. Mapping Major Plant Functional Types using AVIRIS-NG Image at Various Spatial Scales to Aid Wildfire Management in Boreal Alaska presented at 2022 AGU Fall Meeting, Dec. 9-13, Chicago.
17. Waigl, C., Schmidt, J., Stuefer, M., Britz, A., **Panda, S.K.**, Berman, M. and Bhatt, U.S. The 2019 Mckinley Fire In Southcentral Alaska: Burn Severity and Fire Effects from High-Resolution Aerial Imaging Spectroscopy presented at 2022 AGU Fall Meeting, Dec. 9-13, Chicago.
18. Sahoo, S., **Panda, S.K.**, Juday, G.P. 2022. Understanding the climate drivers of forest health and productivity in Alaska using Time-series analysis (2000-2021), presented at the PECORA 2022 conference held in Denver, Colorado during Oct. 23 – 27, 2022.
19. Badola, A. and **Panda, S.K.,** Juday, G.P. 2022. Effect of burn severity on vegetation succession: a case study of 1983 Rosie Creek Fire, interior Alaska, presented at PECORA 2022 conference scheduled held in Denver, Colorado during Oct. 23 – 27, 2022.
20. Waigl, C., Stuefer, M., Badola .A., Smith, W., **Panda, S.K.**, Bhatt, U.S., Waigl, C., Jantd, R. 2022. Boreal forests at tree-scale and stand-scale: Exploring the benefits of in-house aerial VNIR/SWIR imaging spectroscopy for Alaska wildfire research, presented at the 16th International Circumpolar Remote Sensing Symposium held in Fairbanks, Alaska during May 16 – 20, 2022.

21. Badola .A., **Panda, S.K.**, Roberts, D., Bhatt, U.S., Waigl, C., Jantd, R. 2022. Addressing the hyperspectral data needs for the Arctic environment through image simulation, presented at the 16th International Circumpolar Remote Sensing Symposium held in Fairbanks, Alaska during May 16 – 20, 2022.
22. Samuel, W. T., J. A. Falke, **S. K. Panda**, K. D. Tape, A. C. Seitz. 2022. When beavers get burned, do fish pay the price? The role beavers play in mediating the effects of fire on freshwater fish habitat in a boreal ecoregion. Presented at NSF Alaska EPSCoR ("Fire and Ice") All Hands Meeting (2/7/2022); Alaska Chapter American Fisheries Society Meeting 48th Annual (3/2/2022); American Fisheries Society Alaska Chapter, Student Subunit Annual Research Symposium (3/25/2022); Alaska Cooperative Fish and Wildlife Research Unit Annual Meeting (4/6/2022); American Fisheries Society 152nd Annual Meeting (8/21/2022).
23. Waigl, C. Stuefer, M., Badola .A., Smith, C.W., **Panda, S.K.**, Bhatt, U.S., (2021), Quantitative and qualitative insights into boreal forest fire fuel type and condition from tree-scale airborne imaging spectroscopy, presented at 2021 AGU Fall Meeting, Dec. 13-17, New Orleans.
24. Badola .A., **Panda, S.K.**, Roberts, D., Bhatt, U.S., Waigl, C., Jantd, R., (2021), Building a Spectral Library to aid Hyperspectral Data Simulation for Boreal Alaska, [GC15B-0702] presented at 2021 AGU Fall Meeting, Dec. 13-17, New Orleans.
25. Smith, C.W., **Panda, S.K.**, Bhatt, U.S., Meyer, F.J., Badola, A., Hrobak, J., 2021, Assessing wildfire burn severity and its relationship with environmental factors: A case study in the Alaskan boreal forest. 2021 International Boreal Forest Research Association Conference, held (online) during Aug. 16 – 20.
26. Smith, C.W., **Panda, S.K.**, Badola, A., Bhatt, U.S., Meyer, F.J., Haan, C., 2020, Assessing burn severity in the Alaskan boreal forest using remote sensing. American Geophysical Union Fall Meeting 2020, held (online) during Dec. 7 - 11.
27. Badola, A., **Panda, S.K.**, Bhatt U.S., Roberts, D., Smith, C.W., Waigl, C., 2020, Simulating AVIRIS-NG hyperspectral image from Sentinel 2 multispectral image for improved fuel mapping, boreal Alaska. AGU Fall Meeting 2020, held (online) during Dec. 7 - 11.
28. Smith, C.W., **Panda, S.K.**, Bhatt, U.S., Meyer, F.J, Veazy, P., Haan, R., 2019, Imaging spectroscopy of boreal vegetation to aid fire risk assessment, Alaska. AGU Fall Meeting 2019, held during 9 – 13 December in San Francisco, California. Abstract ID: GC51E-1121.
29. Kholodov, A.L., **Panda, S.K.**, Hanson, T., 2018, Establishment of community-based permafrost observation network at the Upper Kuskokwim area as a model of interaction between scientists and indigenous people. European Conference on Permafrost 2018, to be held during June 23 – July 1 in Chamonix, France.
30. Debolskiy, M.V., Nicolsky, D., Romanovsky, V.E., **Panda, S.K.**, and Muskett, R., 2018, Ecotype-based modeling of ground temperature dynamics for Seward Peninsula, Alaska. European Conference on Permafrost 2018, to be held during June 23 – July 1 in Chamonix, France.
31. Payne, C., **Panda, S.K.**, Prakash, A., and Brinkman, T., 2018, Remote sensing of Colville River navigability to aid subsistence travel, Nuiqsut, Alaska. 2018 Alaska Space Grant Program Education and Research Symposium held on April 20, 2018 in Anchorage, Alaska.
32. Kholodov, A.L., **Panda, S.K.**, Hanson, T., Nikolai, T., and Nikolai, S., 2018, Community based observations of permafrost at the Upper Kuskokwim area – the way to empower local communities in the face of environmental changes. 2018 Alaska Forum on Environment Annual Meeting, held during 12 – 16 February in Anchorage, Alaska.
33. **Panda, S.K.**, Kholodov, A.L., Romanovsky, V.E., and Hanson, T., 2017, Integrating remote sensing, field observations, and ground temperature modeling to help address permafrost-related societal challenges around native village of Telida, Interior Alaska, AGU Fall Meeting 2017, held during 11 – 15 December in New Orleans, Louisiana. Abstract ID: C24A-03.

34. Kholodov, A.L., **Panda, S.K.**, and Hanson, T., 2017, Permafrost conditions at the Upper Kuskokwim river area and its influence on local communities, AGU Fall Meeting 2017, held during 11 – 15 December in New Orleans, Louisiana. Abstract ID: GC34C-04.
35. Payne, C., **Panda, S.K.**, Prakash, A., and Brinkman, T.J., 2017, Remote Sensing of Colville River Navigability to Aid Subsistence Travel, Nuiqsut AK, AGU Fall Meeting 2017, held during 11 – 15 December in New Orleans, Louisiana. Abstract ID: EP31D-1895.
36. Romanovsky, V.E., Nicolsky, D., Cable, W., Kholodov, A.L., and **Panda, S.K.**, 2017, New Wave of Permafrost Warming in the Alaskan Interior?, AGU Fall Meeting 2017, held during 11 – 15 December in New Orleans, Louisiana. Abstract ID: GC34C-03 A.
37. Debolskiy, M.V., Nicolsky, D., Romanovsky, V.E., Muskett, R. and **Panda, S.K.**, 2017, Modeling soil temperature change in Seward Peninsula, Alaska, AGU Fall Meeting 2017, held during 11 – 15 December in New Orleans, Louisiana. Abstract ID: C21C-1126.
38. **Panda, S.K.**, Kholodov, A.L., and Hanson, T., 2017, Empowering a remote Arctic community to address climate change impacts on permafrost soil environment, Upper Kuskokwim, Alaska at AMAP International Conference on Arctic Science: Bringing Knowledge to Action, held during 24 – 27 April in Reston, Virginia.
39. Kholodov, A.L., **Panda, S.K.**, Hanson, T., Nikolai, S. Sr., Nikolai, J., Nikolai, S. Jr., Nikolai, A., Ticknor, E., 2017, Organization of the community based permafrost temperature observation network in the Upper Kuskokwim River region, Alaska Forum on Environment Annual Meeting 2017, held during 12 – 16 February in Anchorage, Alaska.
40. **Panda, S.K.**, Kholodov, A.L., and Hanson, T., 2016, Empowering the village communities for sustained observation of permafrost-related environmental changes, Upper Kuskokwim, Alaska, AGU Fall Meeting 2016, held during 12-16 December in San Francisco, California. Abstract ID: GC21F-1161.
41. Whitley, M.A., **Panda, S.K.**, Prakash, A., Brinkman, T.J., 2016, Impacts of Colville River dynamics on river navigability near Nuiqsut, Alaska: 1955-present, AGU Fall Meeting 2016, held during 12-16 December in San Francisco, California. Abstract ID: GC31H-1189.
42. Nicolsky, D., Romanovsky, V.E., **Panda, S.K.**, Marchenko, S.S. and Muskett, R., 2016, Assessment of the permafrost changes in the 21st century under various distrubances to the land cover in the Alaskan Arctic, AGU Fall Meeting 2016, held during 12-16 December in San Francisco, California. Abstract ID: GC31H-1207.
43. Payne, C., **Panda, S.K.**, Prakash, A., Brinkman, T., and Kofinas, G., 2016, Mapping dangerous ice conditions and its impact on subsistence travel on the Colville River, North Slope of Alaska. Poster presented at *Alaska EPSCoR annual meeting* held during Nov. 3-4 at University of Alaska Fairbanks.
44. Sturm, M., **Panda, S.K.**, Cysewski, M., and Cysewski, S., 2016. Documenting change with photos in Nuiqsut. Poster presented at *Alaska EPSCoR annual meeting* held during Nov. 3-4 at University of Alaska Fairbanks.
45. Aggarwal, S., Prakash, A., Balazs, M., **Panda, S.K.**, Waigl, C., Veazey, A., Gens, R., Buzard, R., and Stuefer, M., 2016, Role of data visualization in communicating impacts of climate change and associated risks to communities. Poster presented at *Alaska EPSCoR annual meeting* held during Nov. 3-4 at University of Alaska Fairbanks.
46. Parsekian, A., Schaefer, K., Chen, A., Jafarov, E., Ludwig, S., Liu, L., Michaelides, R., Natali, S., **Panda, S.K.**, Schaefer, S., 2017. Ground penetrating radar investigation of active layer in continuous- and discontinuous- permafrost. *SAGEEP 2017*. Invited talk for Cryosphere Geophysics session.
47. Whitley, M.A., **Panda, S.K.**, Prakash, A., and Brinkman, T.J., 2016, Sixty-five years of Colville river dynamics and its impacts on present river navigability near Nuiqsut, North Slope of Alaska.

*The 14th International Circumpolar Remote Sensing Symposium,* held during 12-16 September, 2016 at Homer, Alaska.

48. **Panda, S.K.**, Romanovsky, V.E., Marchenko, S.S., and Swanson, D.K., 2016, Permafrost in NPS land of Alaska: Past, present and future. *National Park Centennial Symposium,* scheduled to be held during 19-20 October, 2016 at University of Alaska Fairbanks campus.

49. **Panda, S.K.**, Romanovsky, V.E., Marchenko, S.S. and Swanson, D.K., 2016, Past, present and future permafrost distribution in national parks of Alaska: a high-resolution modeling investigation, XI International Conference on Permafrost, scheduled to be held during 20-24 June in Potsdam, Germany.

50. Nicolsky, D., Romanovsky, V.E., **Panda, S.K.**, Muskett, R.R. and Marchenko, S.S., 2016, Applicability of the ecosystem type approach to model permafrost and ground temperature dynamics across the Alaska North Slope, XI International Conference on Permafrost, scheduled to be held during 20-24 June in Potsdam, Germany.

51. Romanovsky, V.E., Nicolsky, D., Marchenko, S.S., Cable, W., Kholodov, A.L., **Panda, S.K.**, and Muskett, R.R., 2016, Past, present and future changes in permafrost temperatures in Alaska, XI International Conference on Permafrost, held during 20-24 June in Potsdam, Germany.

52. **Panda, S.K.**, Kholodov, A., Dubay, C., and Hansen, T., 2016, Engaging rural communities in permafrost and climate monitoring in the Upper Kuskokwim region, interior Alaska, Arctic Observing Summit, held during 15-18 March 2016 in Fairbanks, Alaska. (https://assw2016.org/about/aos)

53. **Panda, S.K.**, Romanovsky, V.E., Marchenko, S.S. and Swanson, D.K., 2015, Forecast of permafrost distribution, temperature and active layer thickness for Arctic National Parks of Alaska through 2100, AGU Fall Meeting 2015, held during 14-18 December in San Francisco, California. Abstract ID: B31D-0591. [1]

54. Nicolsky, D., Romanovsky, V.E., **Panda, S.K.**, Marchenko, S.S. and Muskett, R., 2015, Assessment of the permafrost changes in the 21st century and their impact on infrastructure in the Alaskan Arctic, AGU Fall Meeting 2015, held during 14-18 December in San Francisco, California. Abstract ID: B31D-0615.

55. Romanovsky, V.E., Nicolsky, D., Marchenko, S.S., Cable, W., and **Panda, S.K.**, 2015, Merging field measurements and high resolution modeling to predict possible societal impacts of permafrost degradation, AGU Fall Meeting 2015, held during 14-18 December in San Francisco, California. Abstract ID: B42C-04.

56. Romanovsky, V.E., Cable, W., Kholodov, A., Nicolsky, D., Marchenko, S.S., **Panda, S.K.**, and Muskett, R., 2015, Detecting and forecasting permafrost degradation in a warming climate, AGU Fall Meeting 2015, held during 14-18 December in San Francisco, California. Abstract ID: B31D-0576.

57. Schaefer, K., Chen, A., Liu, L., Parsekian, A., Jafarov, E., **Panda, S. K.**, and Zebker, H., 2015, Realizing full potential of Remotely sensed active layer thickness (ReSALT) products, AGU Fall Meeting 2015, held during 14-18 December in San Francisco, California. Abstract ID: GC23J-1220.

58. Alessa et al., 2015, Best practices for Community Based Observing Networks and Systems (CBONS) – standards, quality assurance, protections and data interoperability, Arctic Observing Open Science Meeting, held during November 17-19, 2015 at Seattle, WA.

59. Schaefer, K., Jafarov, E., Chen, A., Zebker, H., Liu, L., Parsekian, A., and **Panda, S. K.**, 2015, Remotely sensed active layer thickness (ReSALT), 36th Canadian Symposium on Remote Sensing held during June 8-11, 2015 in St. John's, NL, Canada.

60. Schaefer, K., Chen, A., Liu, L., Parsekian, A., Jafarov, E., **Panda, S. K.**, and Zebker, H., 2015, Realizing full potential of Remotely sensed active layer thickness (ReSALT) products, 2015

NASA Carbon Cycle & Ecosystems Joint Science Workshop held during April 20-24, 2015 in College Park Marriott Hotel and Conference Center, Maryland.

61. **Panda, S.K.**, Marchenko, S.S., Romanovsky, V.E., and Swanson, D.K., 2014, Modeling the effects of climate change on permafrost in national parks of Alaska: Will permafrost survive the climate change of 21$^{st}$ century?, AGU Fall Meeting 2014, held during 15-19 December in San Francisco, California. Abstract ID: 10814.
62. Romanovsky, V.E., Cable, W. L., Marchenko, S. S. and **Panda, S.K.**, 2014, Distributed permafrost observation network in western Alaska: the first results, AGU Fall Meeting 2014, taking place during 15-19 December in San Francisco, California. Abstract ID: B33G-04.
63. Romanovsky, V.E., Cable, W. L., and **Panda, S.K.**, 2014, State of permafrost in and around Selawik National Wildlife Refuge, Western Alaska. Western Alaska Interdisciplinary Science Conference, held April 23-25 in Kotzebue, Alaska.
64. **Panda, S.K.**, Marchenko, S.S., Romanovsky, V.E., and Swanson, D.K., 2013, High-resolution near-surface permafrost modeling for the 21$^{st}$ century, Wrangell-St. Elias National Park and Preserve, Alaska, AGU Fall Meeting 2013, held during 9-13 December in San Francisco, California. Abstract ID: 1789949
65. Minsley, B.J., Kass, M.A., Irons, T.P., Bloss, B., Pastick, N., **Panda, S.K.**, Smith, B.D., Abraham, J.D., and Burns, L.E., 2013, A multi-scale permafrost investigation along the Alaska Highway Corridor based on airborne electromagnetic and auxiliary geophysical data, SAGEEP AEM.
66. Minsley, B.J., Kass, M.A., Bloss, B., Pastick, N., **Panda, S.K.**, Smith, B.D., Abraham, J.D., and Burns, L.E., 2012, A multi-scale permafrost investigation along the Alaska Highway Corridor based on airborne electromagnetic and auxiliary geophysical data, AGU Fall Meeting 2012, held December 3-7 in San Francisco, California. Abstract ID: 1503020
67. **Panda, S.K.**, Romanovsky, V.E., Prakash, A., Marchenko, S., and Solie, D.N., 2012, Application of Electromagnetic (EM) Resistivity Data for near-surface permafrost mapping in a pilot study area, interior Alaska, Tenth International Conference on Permafrost, held June 25-29 in Salekhard, Russia.
68. **Panda, S.K.**, Marchenko, S.S., and Romanovsky, V.E., 2012, Geophysical Institute Permafrost Laboratory (GIPL) permafrost dynamics model. Workshop on geospatial data sets for monitoring fires and their effects, held Feb. 15-16 in Fairbanks, Alaska.
69. **Panda, S.K.**, Marchenko, S., Prakash, A. and Romanovsky, V.E., 2010, Modeling of permafrost dynamics at two different biophysical settings near Dry Creek, Interior Alaska. *AGU Fall Meeting 2010*, held December 13 – 17 in San Francisco, California.
70. **Panda, S.K**., 2009b, Remote sensing based near-surface permafrost distribution in a section of the proposed gas pipeline corridor, eastcentral Alaska. At *9$^{th}$ ACUNS International Student Conference on Northern Studies Communities of Change: Building an IPY Legacy*, held October 2-5 in Whitehorse, Yukon, Canada.
71. **Panda, S.K.,** 2009a, Near-surface permafrost mapping using remote sensing and GIS in the Alaska Highway corridor. At the 43rd *Annual Alaska Surveying and Mapping Conference*, held February 24 -25 in Anchorage, Alaska.
72. **Panda, S.K.,** Prakash, A., Solie, D.N., Romanovsky, V.E. and Jorgenson, T.M., 2008b, Identification of surface signatures indicative of near-surface permafrost in the Alaska Highway corridor, *AGU Fall Meeting* 2008, held during 15-19 December in San Francisco, California. [2]
73. Prakash, A., Engle, K., **Panda, S.K.,** Sarkar, S., Margraf, J.F. and Underwood, T., 2008, Nearshore Thermal Habitat and General Circulation Mapping in Arctic Alaskan Coasts using Archived AVHRR images, *AGU Fall Meeting 2008*, held during 15-19 December in San Francisco, California.

***INVITED SEMINARS/ TALKS***

1. **Panda, S.K.** (2026). Gave a research talk 'Geospatial Data Science Natural Resource Applications' at Advanced Indigenous Science and Engineering Society meeting, Brooks Gathering Room, UAF campus, Apr. 9, 2026.
2. **Panda, S.K.** (2025). Gave invited talks during sabbatical leave (Aug. 2024 - May 2025):
    - NIT Rourkela
    - ISER Pune
    - IIT Roorkee
    - IISC Bangalore
    - AWI Potsdam
3. **Panda, S.K.** (2025). Gave an invited talk and participated in the Congressional Briefing, Geospatial Research, Education and Outreach, organized by AmericaView Consortium, Mar. 19th, 2025.
4. **Panda, S.K.** (2023). Gave an invited talk and participated in the panel, Mapping the needs for Alaska's future geospatial workforce at the Alaska GeoSummit held in Anchorage, Oct. 26th, 2023.
5. **Panda, S.K.** (2023). Participated in the panel, New Frontiers in Online Learning: UAF Partnership with edX at the AlaskaX Symposium, Sept. 15th, 2023.
6. **Panda, S.K.** (2023). Advances in Geospatial Data Science Applications delivered in ESS 492/692 Seminar class on Dec. 8th, 2023.
7. **Panda, S.K.** (2022). Geospatial Data Science in Natural Resource Applications, delivered (remotely) on June 18, 2022 at an International Conference on "Geospatial Pathways and Big Data Analytics in Natural Resource Applications and Climate Change" organized by Central University of Jharkhand in association with the Indian Society of Geomatics (ISG), Ranchi Chapter.
8. **Panda, S.K.** (2016). Past, Present and Future permafrost distribution in national parks of Alaska. Delivered on July 27 at 2016 Western Regional Cooperative Soil Survey Workshop, held in Fairbanks, Alaska during July 25-28, 2016.
9. **Panda, S.K.** (2016). Permafrost modeling in Alaska. 2016 International Arctic Research Center Summer School, delivered on July 14, 2016, University of Alaska Fairbanks.
10. **Panda, S.K.** (2016). Permafrost in Alaska: Looking back, Looking ahead. Geophysical Institute Graduate Student Association Seminar Series, delivered on April 11, 2016. University of Alaska Fairbanks.
11. **Panda, S.K.** (2015). Permafrost modeling in Alaska. 2015 International Arctic Research Center Summer School, delivered on May 27, 2015, University of Alaska Fairbanks.
12. **Panda, S.K.**, Marchenko, S.S., Romanovsky, V.E., and Swanson, D.K. (2014). Fate of permafrost in Denali National Park and Preserve – A modeling investigation. International Arctic Research Center Seminar, delivered on February 27, 2014.

***IN NEWS ARTICLES AND MEDIA***

1. AFES Faculty Research Profile: Mapping Alaska with Santosh Panda
2. I appeared as a guest on the Spatial Signal podcast, hosted by Dr. Brad Shellito and Dr. Aaron Maxwell, Everything was going fine and then (Jan. 11th, 2026)
3. Melting permafrost may make oil production nearly impossible on the North Slope, by Tim Bradner. Published in Anchorage Press on 11/27/2019.
4. Alaska oil, gas exploration threatened by thawing permafrost: Fuel for Thought. Published on 10/29/2019.

5. Permafrost and Community, PolarTREC informal educator Allyson Woodard's blogs about her field experience in Nikolai, Alaska during Aug. 10 – 18, 2018, published on PolarTREC site.
6. *Intersection of indigenous communities and modern research*, published in Polarfield blog.
7. *Geophysical Institute creates travelling permafrost exhibit*, published in Fairbanks Daily News-Miner.
8. *University brings permafrost science to the public*, published in The Sun Star.
9. Participated in a Press Conference on 'Alaska's thawing permafrost: Latest results and future projections' at the American Geophysical Union Fall meeting in San Francisco, held on Dec. 15, 2015. Recording: https://www.youtube.com/watch?v=49kqqPIrNQA
10. *This is what happens when the Arctic warms twice as fast as the rest of the planet*, published in The Washington Post. Available at: http://tinyurl.com/arcticreport2015
11. *North Slope permafrost thawing sooner than expected* published on UAF News.
12. *North Slope permafrost thawing more quickly than expected* published on Alaska Dispatch News.
13. **Panda, S.K.**, Schaefer, K., Liu, L., Jafarov, E., Parsekian, A.D., and Chen, A., 2015, Linking lake area change to ground subsidence in Prudhoe Bay, North Slope of Alaska. *Changing Ice Newsletter*.
14. Most of Alaska's permafrost could thaw this Century. Available at http://www.livescience.com/49184-permafrost-disappears-from-alaska.html
15. **Panda, S.K.**, 2014, Permafrost Tunnel outreach project receives $1.5 million grant from National Science Foundation. *Geophysical Institute Newsletter*, Sept. 29, 2014. http://newsletter.gi.alaska.edu/node/4538

**AWARDS AND SCHOLARSHIPS**

- 2025 R1 Faculty Mentorship Grant Writing Award of $14,800. This award is to support writing of a grant that includes funding support for a UAF PhD student.
- 2025, Outstanding Service award from AmericaView. This award acknowledges my support for the organization and my contributions to Earth observation education. AmericaView is a nationwide, university-based, and state-implemented network that advances Earth observation education through remote sensing science, applied research, workforce development, technology transfer, and community outreach.
- 2023 edX prize finalist for innovation and exceptional contributions in online teaching. The finalists represent faculty and educators from 250 institutions around the globe behind a variety of Massive Open Online Courses (MOOCs), ranging in topics from healthcare to sustainability and mathematics.
- 2023 Faculty Fellowship Award of $10,000 from UAF Faculty Accelerator Program. This award is for a summer project that takes advantage of the university's research and expertise in addressing a local community-based problem with tangible impact on the community.
- 2023 NSF Alaska EPSCoR Fall Travel Award of $3,500 to attend 2024 American Associations of Geographers annual meeting held in Honolulu, Hawaii during April 16 - 20, 2024.
- 2023 NSF EPSCoR Travel Award of $3,500 to attend 2023 International Geoscience and Remote Sensing Symposium held in Pasadena, California during July 16 - 21, 2023.
- 2022 Certificate of Appreciation for an invited talk on 'Geospatial Data Science in Natural Resource Applications', delivered (remotely) on June 18, 2022 at an International Conference on "Geospatial Pathways and Big Data Analytics in Natural Resource Applications and Climate

Change" organized by Central University of Jharkhand in association with the Indian Society of Geomatics (ISG), Ranchi Chapter.

- 2021 William T. Pecora Group Award: the AmericaView consortium is recognized for advancing Earth observation education through remote sensing science, applied research, workforce development, technology transfer, and community outreach. Sponsored by the U.S. Geological Survey (USGS) and the National Aeronautics and Space Administration (NASA), the annual award has been presented since 1974 and honors the memory of Dr. William T. Pecora, former USGS Director and Department of the Interior Undersecretary.
- U.S. Globe Change Research Program Travel Award to participate in AMAP International Conference on Arctic Science: Bringing Knowledge to Action in Reston, VA during April 24 – 27, 2017.
- Travel award from Alaska EPSCoR to participate in NSF Day Workshop in Anchorage, Alaska on April 8, 2016.
- Make an Impact: The Arctic in the Classroom (TAC) Workshop 2016 held during the Arctic Science Summit Week 2016 at University of Alaska Fairbanks campus during Mar. 13-15, 2016, organized and sponsored by Arctic Research Consortium of the United States.
- United States Permafrost Association's Travel award to participate in the 2015 American Geophysical Union Fall Meeting held during Dec. 14-18, 2015 in San Francisco, CA.
- Travel award to participate in Community Based Observing Networks and Systems (CBONS) held during Oct. 5-6, 2015 at University of Washington campus, Seattle, WA; organized by the Center for Resilient Communities at the University of Idaho.
- Travel award to participate in New Generation of Polar Researchers (NGPR) Leadership Symposium held during May 2-9, 2015 at USC Wrigley/Boone Center for Environmental Leadership, Catalina Island, CA; sponsored by the U.S. National Science Foundation Division of Polar Programs. (http://disccrs.org/ngpr)
- Travel award to participate in 2015 ENGAGE Workshop held during January 18-20, 2015 in Arlington, VA. Organized by Incorporated Research Institutions for Seismology (IRIS) and sponsored by the U.S. National Science Foundation.
- 2014 Arthur J. Schaible Memorial Cultural Enrichment Opportunity award of $1700 for a Career Development Workshop at the University of Alaska Fairbanks. (Co-PI)
- 2012 Young Researcher Stipend from Government of Yamal-Nenets autonomous district to participate in the Tenth International Permafrost Conference, in Salekhard, Yamal-Nenet Peninsula, Russia.
- Travel Support from SVALASKA project to participate in "Permafrost modeling in the Nordic region" course held in the Department of Geoscience, University of Oslo, Norway, during June 10-22, 2012.
- 2010 Global Change Student Research Grant of $8,000 from Center for Global Change and Arctic System Research, International Arctic Research Center (University of Alaska Fairbanks). Project title: Modeling of permafrost dynamics along Alaska Highway corridor, Interior Alaska. (PI)
- Graduate School Dissertation and Thesis Completion Award 2010-11, University of Alaska Fairbanks. Award: 18 Tuition Credits + $13,500 Stipend.
- 2010 IARC Summer School Scholarship to participate in an interdisciplinary summer course on 'Arctic in a changing climate: Physical and biological linkage to permafrost', held at the University of Alaska Fairbanks, May 20 – June 5, 2010.
- PYRN-USPA Best Poster Presentation Award at the American Geophysical Union General Fall Meeting held during December 2008 at San Francisco, California.

- Alaska Space Grant Program: Special Research Project Fellowship Recipient, 2007-2009, $40,000.
- Alaska Division of Geological and Geophysical Surveys: Research Fellowship for Gas Pipeline Project, 2007-2009, $40,000.
- National Doctoral Fellowship from All India Council of Technical Education, New Delhi, for conducting research at the Indian Institute of Technology Roorkee, India during 2006.
- Ministry of Human Resources and Development (MHRD) Institute Fellowship for conducting research at the Indian Institute of Technology Roorkee, India during 2005.
- Council of Scientific and Industrial Research Internship award for conducting research at the Central Mining Research Institute, Dhanbad, India during 2004.
- 2001 Outstanding Student Achievement award for securing 1st Rank in the University in Geology, Sambalpur University, Odisha, India.

**PROFESSIONAL DEVELOPMENT** (TRAINING/WORKSHOPS)

1. 2026, attended a workshop on Practical Pathways to Ungrading and Authentic Assessment by Dr. Jesse Stommel, Apr. 14, 2026.
2. 2026, attended a webinar on Managing Your Emotions in the Workplace by ComPsych-Guidance Resources on Mar. 20, 2026.
3. 2026, attended a webinar on The Art of Patience by ComPsych-Guidance Resources on January 15th, 2026.
4. 2025, attended a webinar on Parental Burnout: How to overcome challenges and thrive through parenthood by ComPsych-Guidance Resources on June 27th, 2025.
5. 2025, Completed Research Security Training offered online by CITI Program. Completion Date: June 25, 2025.
6. 2025, Civil Rights Training by Alda Norris, IANRE
7. 2025, attended webinar Psychological Avoidance and its Impact on your Mental Health by ComPsych-Guidance Resource on Mar. 27, 2025.
8. 2025, attended webinar Managing Worry and Anxiety offered by ComPsych-Guidance Resource on Mar. 6, 2025.
9. 2024-2025, A NatureCareer podcast series on 'Mental health in academia' by Adam Levy.
10. 2024, attended webinar How to support well-being with emotional intelligence offered by Hazelden Betty Ford Foundation on Oct. 16, 2024.
11. 2024, attended 28th National EPSCoR conference organized by Nebraska EPSCoR and the NSF held (10/13 - 10/16) in Omaha, Nebraska.
12. 2024, participated in 2024 Geo for Good Earth Engine Summit hosted by Google Inc during Sept. 23 – 26 in Dublin, Ireland.
13. 2024, participated in a 3-day Unmanned Aerial Systems training led by Jeremy Crowley, University of Montana Autonomous Aerial Systems Office held (8/16 – 8/18, 2024) at UAA campus.
14. 2024, attended Becoming a Better Listener webinar offered by ComPysch on July 19, 2024.
15. 2023, participated in Micro what? All things microcredentials and digital badging by UAF CTL staff Faith Feagle on Oct. 24th, 2023.
16. 2023, participated in Canvas Foundation: Data and Analytics training conducted by a Canvas Instructor on Oct. 18th, 2023.

17. 2023, participated in Canvas Foundation: Grading and Feedback training conducted by a Canvas Instructor on Oct. 12th, 2023.
18. 2023, participated in Cultural Humility Workshop offered by Dr. Cana Uluak Itchuaqiyaq held on Sept. 20th, 2023. The workshop was focused on conducting equitable research with Arctic and Indigenous communities.
19. 2023, participated in Quality Matters Workshop: Improving Your Online Course offered by UA Instructional Designers (Katie Walker and Dan LaSota) office held (Aug. 14-15, 2023).
20. 2023, attended Tools to Handle Stress webinar offered by ComPysch on July 28, 2023.
21. 2023, attended Using Reason to Resolve Conflict webinar offered by ComPysch on June 16, 2023.
22. 2023, participated in a 1.5-day workshop on Indigenous Ways and Western Science presented by Ilarion Merculieff and Libby Roderick held (May 10-11) on UAF Campus. This workshop was the first event on the UAF Moving Forward Together Project.
23. 2023, participated in a 3-day writing and storytelling workshop by Seth Kanter organized by MIX at UAF (Mar. 6 - 8, 2023).
24. Completed a certification course on social annotation in teaching from Hypothesis Academy held during Jan. 3 - 17, 2023.
25. 2022, Completed an online short-course on Mapping and Data Visualization with Python offered by Spatial Thoughts, 12/19 – 12/22, 2022.
26. Participated in CNSM Strategic Planning Workshop held during Nov. 1 - 2, 2022.
27. The (virtual) "Write Winning Grant Proposals" workshop led by John D. Robertson of Grant Writers' Seminars and Workshops, held during May 23 – 25, 2022.
28. Participated in 2019 NSF Engineering CAREER proposal writing workshop during April 1 – 2, 2019 held in Arlington, VA.
29. Heartsaver CPR AED training from Interior Region EMS Council, completed on April 4, 2019.
30. Alaskan Gatekeeper Suicide Prevention Training: A QPR Approach completed on Sept. 10, 2018. Sponsored by the State of Alaska, Department of Health and Social Services, Behavioral Health, Prevention and Early Intervention Services.
31. Bear Safety training from Tim Craig; completed on: 3/7/2019; 8/3/2018.
32. Alaska National Lab Day 2018: Exploring research collaboration between the University of Alaska and U.S. Department of Energy national laboratories, held during May 30 – 31, 2018 at the University of Alaska Fairbanks Campus.
33. Workshop on the Application of smallsat and commercial imagery to Antarctic Science held during May 21 - 22, 2018 at the University of Minnesota, Minneapolis Campus, MN; organized by the Polar Geospatial Center and sponsored by the National Science Foundation.
34. Responsible Research Conduct workshop offered by UAF BLaST program on April 8, 2017.
35. Arctic Winter Field training and Snowmachine operation training by CPS Polar Field Services. Mar. 15, 2017. Instructor: Matt Iringa.
36. Participated in a FLC book discussion as part of the Faculty Development program led by Joy Morrison. Period: October – December, 2016; 1-hour meeting twice a month.
37. NSF Day Workshop held on April 8, 2016 at University of Alaska Anchorage, organized by National Science Foundation.
38. Make an Impact: The Arctic in the Classroom (TAC) Workshop 2016 held during Mar. 13-15, 2016 at the Arctic Science Summit Week 2016, University of Alaska Fairbanks, organized and sponsored by Arctic Research Consortium of the United States.

39. Fall 2015 Alaska Fire Science Workshop held on Oct. 16, 2015 at Bureau of Land Management, Fairbanks, AK.
40. Workshop on Community Based Observation System and Network held during Oct. 5-6, 2015 at University of Washington, Seattle, WA; organized by the Center for Resilient Communities at the University of Idaho.
41. Student mentoring training for faculty held on Aug. 24, 2015 at University of Alaska Fairbanks.
42. New Generation of Polar Researchers (NGPR) Leadership Symposium held during May 2-9, 2015 at USC Wrigley/Boone Center for Environmental Leadership, Catalina Island, CA; sponsored by the U.S. National Science Foundation Division of Polar Programs. (http://disccrs.org/ngpr)
43. 2015 ENGAGE Workshop held during January 18-20, 2015 in Arlington, VA. Organized by Incorporated Research Institutions for Seismology (IRIS) and sponsored by the U.S. National Science Foundation.
44. Collaborative Institutional Training Initiative (CITI Program) on Social Behavioral Research – Human Subjects. Completed on 6/18/2015; ID: 4890825. Expiration Date: 6/17/2018
45. Wilderness and Remote First Aid, Adult CPR/AED, and Bloodborne Pathogens Training from LTR Training Systems, Inc.: completed on Mar. 10, 2015.
46. Arctic Field Training from Polar Field Services, Mar. 25, 2015.
47. Bear Safety and Shotgun Instruction from Joe Nova: Completed on April 15, 2008; Mar. 19, 2014; Mar. 22, 2016.

**FIELD EXPERIENCES** (WORK/TRAINING)

1. Tree measurements and tree mortality surveys within Bonanza Creek Experimental Forest. Summer 2022-2024.
2. Wildfire burn severity surveys of 2019 Shovel Creek fire and 2019 Nugget Creek fire. Summer 2020.
3. Vegetation composition survey within Bonanza Creek Experimental Forest, Caribou-Poker Research Watershed, and UAF North Campus. Summer 2018 - 2022.
4. Snow survey and coring around Telida village, Upper Kuskokwim, Alaska. Mar. 22 – 31, 2017.
5. Survey of ecotype and seasonal thaw depth, soil sampling, installation of ground temperature and surface weather monitoring stations around Telida and Nikolia villages, Upper Kuskokwim, Alaska. Aug. 2016, 2017, and 2018.
6. Spectral measurements of river water, and water depth along a 50-mile stretch of the Colville River, near Nuiqsut, Alaska. Aug. 2016 and 2017.
7. Vegetation survey and seasonal thaw depth survey near Teller and Kougrak, Seward Peninsula, Alaska.
8. GPR survey of seasonal thaw depth along six transects spread across 100s of miles in the North Slope of Alaska.
9. Survey of seasonal thaw depth, retrieval of data from permafrost temperature monitoring stations along the Dalton Highway, Alaska. Summer 2013, 2014, and 2015.
10. Installation of 20 long-term ground temperature monitoring stations, data retrieval, and maintenance along the Richardson and the Alaska Highway. Summer 2014 – 2016.
11. 2012 Friends of Pleistocene field excursion on geology and geo-hazards along Alaska Highway and Glenn Highway led by Dr. Richard (Dick) Reger, Dr. Trent Hubbard, and Dr. Patricia Gallagher.

12. Field work focused on Arctic vegetation, soil, and permafrost along the Dalton Highway as a part of 2010 IARC Interdisciplinary Summer School on '*Arctic in a changing climate: Physical and biological linkage to permafrost*' led by Dr. Donald Walker and Dr. Ronald Daanen, June 2010.
13. Snow survey along the Alaska Highway. Mar. 2008 and 2009.
14. Survey of seasonal thaw depth, permafrost presence/ absence, soil properties, and installation of ground temperature monitoring stations along Alaska Highway. Summer 2007 – 2011.
15. 2009 Friends of Pleistocene field excursion around Fairbanks on paleoecology, periglacial geomorphology, fire history and ecology, and permafrost features led by Dr. Nancy Bigelow and Ms. Patty Burns. Fall 2009.
16. Local field site visits around Fairbanks, Delta Junction, and Livengood as part of a class on General Geology of Alaska led by Dr. Rainer Newberry. Spring 2009.
17. Local field site visits around Fairbanks as part of Ninth International Permafrost Conference. Summer 2008.
18. 2008 Friends of Pleistocene field excursion in Juneau on history of Cordilleran glaciation, Neogene extensional volcanism, active strike slip faulting, dynamic sea level change, early Holocene and post LIA uplift of marine sediments, led by Dr. Cathy Connor. Fall 2008.
19. Glaciology fieldwork at Gulkana glacier as part of a glaciology field course. Summer 2008.
20. 2007 Friends of Pleistocene field excursion on glaciations and geologic history of Kenai Peninsula, Alaska led by Dr. Richard (Dick) Reger, Dr. Ed Berg, and Ms. Patty Burns. Fall 2007.
21. Field training on economic geology, open pit marble and sandstone mine visits in Rajasthan, as part of MS graduate course work. Spring 2003.
22. Field training on structural and geologic mapping in the foothills of Himalaya around Dharamshala, Himachal Pradesh, as part of MS graduate course work. Spring 2002.
23. Field training on geologic mapping, coastal processes, coastal geomorphology, and visit to open pit iron mine around eastern coastal Odisha as part of undergraduate field course. Spring 2001.
24. Field visit to underground pb-zn Sargipali mine, open pit Gopalpur coal mine, Sundargarh, Odisha. Fall 2000.